An Affine Residual Enclosure for a Projected Resolvent Functional

Jongmin Choi
Independent Researcher, Seoul, Korea
24ping@naver.com
ORCID: 0009-0008-7448-514X

Abstract

For a positive self-adjoint operator, we record an affine residual enclosure for a scalar projected resolvent functional. If $A \geq \lambda \,\mathrm{Id}$, a trial vector Ex_0 has residual $r_0 = g - AEx_0$, and $l = Em$, the center contains the essential correction $\langle l, r_0\rangle/(2\lambda)$ and the radius is $\|m\|\|r_0\|/(2\lambda)$. This follows directly from the standard operator-order estimate $\|2\lambda A^{-1} - \mathrm{Id}\| \leq 1$; no claim of bibliographical priority is made. Exact Galerkin orthogonality removes the correction, whereas an inexact trial generally does not. We also prove that the enclosure at $x_0 = 0$ always contains zero. A concrete application is specified for an auxiliary operator on a finite interval, with the complete exterior potential and the actual prime shifts $2, 3, 4$. Its canonical weighted graph reserve is reconstructed and distinguished from an unweighted row sum. An analytic lower bound is proved from explicit identities and elementary estimates. The required scalar moments and full residual norms have not been numerically certified for the fixed source. No sign certification or conclusion about the Riemann hypothesis is obtained.

1 Purpose and scope

Residual bounds for positive self-adjoint operators are standard consequences of operator order and spectral calculus; see, for example, the framework in [1, Chapter 2]. Here the purpose is to specify a scalar target, retain the correction required by an inexact trial, and make the source definitions of a concrete finite-interval application explicit.

The abstract result is independent of any arithmetic or prime-shift model. The concrete application uses an auxiliary operator A_b on $L^2_{\mathrm{even}}((-3/4, 3/4); \mathbb{R})$. It is not the original operator on the whole real line, and no identification with an original minimum Gamma realization is asserted. The manuscript contains analytic proofs and a definition audit. It does not present a newly evaluated response, Gram matrix, quadrature, or signed numerical enclosure.

2 The affine residual enclosure

Let $\mathcal{H}$ be a real Hilbert space. Let A be a densely defined self-adjoint operator with $A \geq \lambda \,\mathrm{Id}$ for a fixed $\lambda > 0$. Thus A^{-1} is a bounded positive self-adjoint operator on $\mathcal{H}$. Let $E : \mathbb{R}^d \to \mathcal{H}$ be an isometry with $\operatorname{Ran} E \subset \mathrm{D}(A)$. Fix $F \in \mathcal{H}$, $a \in \mathbb{R}$, and $\alpha, m \in \mathbb{R}^d$, with $E^*F = 0$, and define

$$g = aF - E\alpha, \qquad G = E^*AE, \qquad l = Em, \qquad M = \langle l, A^{-1}g\rangle. \tag{2.1}$$

The adjoint E^* and all norms refer to the Hilbert structure in question. For any fixed $x_0 \in \mathbb{R}^d$, set

$$r_0 = g - AEx_0, \qquad \delta = m^T(-\alpha - Gx_0). \tag{2.2}$$

Because $E^*g = -\alpha$, one has $E^*r_0 = -\alpha - Gx_0$ and $\delta = \langle Em, r_0\rangle$.

Theorem 2.1 (Affine residual enclosure)**.** *Under these assumptions,*

$$\boxed{\left|M - m^T x_0 - \frac{\delta}{2\lambda}\right| \le \frac{\|m\|\|r_0\|}{2\lambda}.} \tag{2.3}$$

In particular the residual norm in this statement is the full norm in $\mathcal{H}$, not the norm of its projection $E^* r_0$.

Proof. The spectral theorem gives $0 \le A^{-1} \le \lambda^{-1}\,\mathrm{Id}$, and hence

$$\|2\lambda A^{-1} - \mathrm{Id}\| \le 1, \qquad \|A^{-1} - (2\lambda)^{-1}\,\mathrm{Id}\| \le (2\lambda)^{-1}. \tag{2.4}$$

Since $Ex_0 \in \mathrm{D}(A)$, $A^{-1}g = Ex_0 + A^{-1}r_0$. The isometry of E therefore yields

$$M - m^T x_0 - \frac{\delta}{2\lambda} = \langle Em, (A^{-1} - (2\lambda)^{-1}\,\mathrm{Id})r_0\rangle.$$

Apply Cauchy–Schwarz and (2.4), using $\|Em\| = \|m\|$. □

Corollary 2.2 (Exact Galerkin trial)**.** *The matrix G is symmetric and $G \ge \lambda I_d$. The unique solution $x_* = -G^{-1}\alpha$ satisfies $E^*(g - AEx_*) = 0$. Thus*

$$\left|M - m^T x_*\right| \le \frac{\|m\|\|g - AEx_*\|}{2\lambda}. \tag{2.5}$$

For a general inexact x_0, the correction in (2.3) must be retained.

Proof. For $y \in \mathbb{R}^d$, $y^T G y = \langle Ey, AEy\rangle \ge \lambda\|y\|^2$. Symmetry follows from self-adjointness and $\operatorname{Ran} E \subset \mathrm{D}(A)$. The remaining assertions follow from $E^* r_0 = -\alpha - Gx_0$. □

Remark 2.3 (Comparison with the usual residual bound). The ordinary estimate is $|M - m^T x_0| \le \|m\|\|r_0\|/\lambda$. For the same full residual and the same λ, (2.3) has half that radius, but generally a different center. Its exact interval is contained in the ordinary interval because $|\delta| \le \|m\|\|r_0\|$. If δ, m, or r_0 are themselves enclosed imperfectly, their additional errors must also be included. No factor-of-two improvement in an actual stored numerical certificate follows from this comparison alone. The constant in (2.4) is the sharp uniform constant, already attained by $A = \lambda\,\mathrm{Id}$.

Proposition 2.4 (Zero-start limitation)**.** *For $x_0 = 0$, the norm-based interval in (2.3) always contains zero. Consequently this interval alone cannot certify a strict sign of M from the zero trial.*

Proof. Here $r_0 = g$, $m^T x_0 = 0$, and $\delta = \langle Em, g\rangle$. Cauchy–Schwarz gives $|\delta| \le \|m\|\|g\|$. The absolute value of the center $\delta/(2\lambda)$ is therefore no larger than the radius $\|m\|\|g\|/(2\lambda)$. □

The proposition does not assert $M = 0$. To obtain a sign by this mechanism, one needs a nonzero useful trial or additional information beyond this general norm interval. It supplies no reason to start a zero-trial numerical experiment merely to certify a sign.

3 The finite-interval source operator

Fix

$$B = \frac{3}{4}, \quad \Omega = (-B, B), \quad \theta = \frac{1}{65536}, \quad \omega = 1 - \theta, \quad a = \frac{261572074620463589}{296898235056783360} > 0. \tag{3.1}$$

Let $t = \log 2$, $u = \log 3$, and

$$c_2 = \frac{t}{\sqrt{2}}, \qquad c_3 = \frac{u}{\sqrt{3}}, \qquad c_4 = \frac{t}{2}.$$

Every function on Ω is extended by zero outside it. If $T_s f(x) = \widetilde{f}(x-s)$ for $x \in \Omega$, define

$$\mathcal{P}_B = c_2(T_t + T_{-t}) + c_3(T_u + T_{-u}) + c_4(T_{2t} + T_{-2t}). \tag{3.2}$$

This is a bounded self-adjoint operator, preserves evenness, and retains both directions of the shifts. The coefficient of the shift $\log 4 = 2t$ is $\log 2/2$, with no additional duplicate prime-power term.

For $s > 0$ put

$$\rho(s) = \frac{e^{-s/2}}{1-e^{-2s}}, \qquad R(s) = \int_s^\infty \rho(v)\,\mathrm{d}v = \operatorname{artanh}(e^{-s/2}) + \arctan(e^{-s/2}), \tag{3.3}$$

$$V(x) = R(B+x) + R(B-x), \qquad V_0 = 2R(B), \tag{3.4}$$

$$c_\Gamma = \log\pi + \gamma_E + \frac{\pi}{2} + 3\log 2, \qquad \kappa_g = -c_\Gamma + \frac{17}{5}\omega + 7\theta. \tag{3.5}$$

Here γ_E is Euler's constant. The tail in (3.3) is complete. The two terms in (3.4) are the two exterior costs associated with zero extension; neither is truncated or multiplied by two again.

On $\mathcal{H}_b = L^2_{\mathrm{even}}(\Omega;\mathbb{R})$ define the auxiliary operator

$$A_b = (a+\kappa_g)\,\mathrm{Id} + \omega M_V - \mathcal{P}_B, \qquad \mathrm{D}(A_b) = \{f \in \mathcal{H}_b : Vf \in L^2(\Omega)\}, \tag{3.6}$$

where M_V is multiplication by V. Its form domain is

$$\mathcal{V}_b = \left\{ f \in \mathcal{H}_b : \int_\Omega V|f|^2\,\mathrm{d}x < \infty \right\}. \tag{3.7}$$

The real multiplication operator, plus a bounded self-adjoint perturbation, is self-adjoint on the displayed operator domain. Its semibounded closed form has domain (3.7). The positive lower bound will be proved in Section 5. In (3.6), ω multiplies the potential; the off-diagonal prime coefficients are $-c_2, -c_3, -c_4$, not $-\omega c_j$ and not $-2c_j$. The constant κ_g also depends on ω through (3.5).

3.1 Physical basis and the fixed polynomial data

Use the physically orthonormal even Legendre functions

$$e_{2j}(x) = \sqrt{\frac{4j+1}{2B}}\,P_{2j}(x/B), \qquad 0 \le j < 16, \tag{3.8}$$

and let $Ey = \sum_{j=0}^{15} y_j e_{2j}$ and $P = EE^*$. The normalization follows from Legendre orthogonality; see [2, Section 18.3]. The fixed high polynomial is specified by rational coefficients b_m as

$$F(x) = \frac{1}{\sqrt{2B}} \sum_{m=32,34,\ldots,128} b_m P_m(x/B). \tag{3.9}$$

There are 49 coefficients. The exact fixed instance and the rational vector $\alpha \in \mathbb{R}^{16}$ are preserved as source strings in `source_definitions.json`; no decimal endpoint transcription is used to define them. The results here hold for any such fixed coefficients. The archived instance is identified by the source hashes in Section 7, and is not asserted to be an exact full inverse merely because it arose as a finite inverse prefix.

The degree separation gives $E^*F = 0$. The endpoint growth $V(x) = O(1 + |\log(B - |x|)|)$ follows from (3.3). Since $\int_0^\varepsilon |\log s|^2\,\mathrm{d}s < \infty$, every e_{2j} and F belongs to $\mathrm{D}(A_b)$. This establishes the operator membership needed for $g - A_b E x_0$, which is stronger than mere form membership. It does not prove membership in any original whole-line minimum operator domain.

4 The canonical weighted graph reserve

4.1 Complete component geometry

Put $r_g = B - t$, $v_g = u - t$, and $w_g = 2t - u$. Up to null endpoints, the graph of the shifts in (3.2) is described by three component charts:

$$\begin{aligned} J_7 &= (-r_g, r_g), & \xi_7(x) &= (x, x+t, x-t, x+v_g, x-v_g, x+w_g, x-w_g),\\ J_4 &= (u+B-2t, 2t-B), & \xi_4(y) &= (y-t, y, y-u, y-u+t),\\ J_2 &= (B+t-u, u-B), & \xi_2(z) &= (z-t, z). \end{aligned}$$

All coordinate Jacobians are one. With zero-based vertex labels, the edges are

Chart	c_2 edges	c_3 edges	c_4 edges
7	$(0,1),(0,2),(4,5),(3,6)$	$(1,4),(2,3)$	$(1,2)$
4	$(0,1),(2,3)$	$(1,2)$	none
2	$(0,1)$	none	none

For completeness, the positive physical half is partitioned by the seven bands in Table 1. Their vertex images are disjoint except at endpoints. The negative half is obtained by reflection: the chart parameters map as $x \mapsto -x$, $y \mapsto u - y$, and $z \mapsto t - z$, with the corresponding vertex permutations. Checking the translated coordinates against $\pm B$ gives exactly the edge table above. In particular both central $n = 2$ neighbors and the $n = 4$ edge occur.

The inequalities needed for this partition follow from the positive series

$$2\sum_{j=0}^{N-1}\frac{q^{2j+1}}{2j+1} < \log\frac{1+q}{1-q} < 2\sum_{j=0}^{N-1}\frac{q^{2j+1}}{2j+1} + \frac{2q^{2N+1}}{(2N+1)(1-q^2)}, \quad 0 < q < 1. \tag{4.1}$$

Using $(q, N) = (1/3, 3), (1/2, 4)$ gives $693/1000 < t < 694/1000$ and $1098/1000 < u < 1099/1000$. These bounds imply $0 < r_g < w_g < v_g < t < B$, $2u - t > 2B$, $4t - u > 2B$, $t + u > 2B$, and $3t - u > B$. Also $\log 5 > 3/2$, from the first three positive terms in (4.1) with $q = 2/3$. Thus shifts with $n \geq 5$ have no positive-measure overlap on this support.

Translation of representatives therefore gives the unitary identification

$$L^2(\Omega) \simeq L^2(J_7; \mathbb{R}^7) \oplus L^2(J_4; \mathbb{R}^4) \oplus L^2(J_2; \mathbb{R}^2).$$

This is an assertion about almost-everywhere representatives, not point evaluation of an abstract L^2 class. It restricts to the reflection-even reducing subspace. Each undirected edge contributes $2c_{ij}f_if_j$, representing both arithmetic directions once.

4.2 Weighted rows and the reserve function

For positive weights w_i, the exact edge identity is

$$c_{ij}\left(\frac{w_j}{w_i}|z_i|^2 + \frac{w_i}{w_j}|z_j|^2 - 2z_iz_j\right) = c_{ij}w_iw_j\left|\frac{z_i}{w_i} - \frac{z_j}{w_j}\right|^2. \tag{4.2}$$

Consequently the diagonal associated with these squares is the *weighted* actual row $d_{\mathrm{act},i} = \sum_{j\sim i} c_{ij}w_j/w_i$.

The elementary bounds $c_2 < 1/2$, $c_3 < 13/20$, $c_4 < 7/20$ follow from the logarithm bounds above and $1/\sqrt{2} < 71/100$, $1/\sqrt{3} < 7/12$. They are used only for inequalities, not to redefine (3.2). Choose reserve weights $(10/9, 1, 9/10, 1/2)$ for the seven chart's center, t, v, and w vertex types. The four weighted row bounds are

$$\frac{9}{10}, \qquad \frac{5}{9} + \frac{117}{200} + \frac{7}{20} = \frac{2683}{1800}, \qquad \frac{13}{18} + \frac{5}{18} = 1, \qquad \frac{9}{10}.$$

For the path choose weights 3/5 at leaves and 1 at interior vertices; the bounds are 5/6 and 19/20. For the pair use equal weights and the bound 1/2. This defines the bounded even function $\overline{d}$ in the table.

Positive physical band	Vertex type	$\overline{d}$
$(0, r_g)$	seven center	9/10
$(r_g, 3t-u-B)$	path leaf	5/6
$(3t-u-B, B+t-u)$	seven w	9/10
$(B+t-u, u-B)$	pair	1/2
$(u-B, u+B-2t)$	seven v	1
$(u+B-2t, 2t-B)$	path interior	19/20
$(2t-B, B)$	seven t	2683/1800

Table 1: The canonical reserve on Ω. Use even reflection on the negative half; values at the finitely many boundaries are immaterial.

Summing (4.2) gives, on finite weighted energy vectors,

$$S_{\mathrm{graph}}[f] := \int_\Omega \overline{d}|f|^2 \,\mathrm{d}x - \langle f, \mathcal{P}_B f\rangle \geq 0. \tag{4.3}$$

Its exact decomposition retains the weighted edge squares and the nonnegative diagonal slack $\overline{d} - d_{\mathrm{act}}$.

Proposition 4.1 (The reserve is not an unweighted row sum)**.** *The definition $\overline{d} = \mathcal{P}_B 1_\Omega$ is incompatible with Table 1. It changes the multiplier $p_\eta = \omega(W_B + 9/10 - \overline{d})$ and any moment derived from it. In fact that replacement makes this multiplier negative near the center.*

Proof. On the central band only the two $n = 2$ shifts meet the constant supported function, so $\mathcal{P}_B 1_\Omega = 2c_2 = \sqrt{2}\log 2$, whereas the canonical reserve is 9/10. Since $\log 2 > 2/3$ and $\sqrt{2} > 7/5$, $\sqrt{2}\log 2 > 14/15 > 9/10$. At zero, $W_B(0) = 0$, and continuity on the central band proves the asserted negativity on a positive-measure neighborhood. □

The distinction also applies to d_{act}, which is weighted, not $\mathcal{P}_B 1_\Omega$. Replacing $\overline{d}$ consistently in an algebraic decomposition does not alter (3.6), but it alters p_η, m, and the scalar target. A bounded even change of $\overline{d}$ leaves the relevant weighted domain equivalence intact; it does not preserve $p_\eta \geq 0$ or the numerical functional.

5 An explicit analytic lower bound

We reconstruct the estimates required for a lower bound. No stored audit status or numerical response output is used as a proof premise.

Lemma 5.1 (Complete logarithmic reserve)**.** *For $x \in \Omega$,*

$$V(x) \geq V_0 + W_B(x), \qquad W_B(x) = -\frac{1}{2}\log(1 - x^2/B^2).$$

Moreover $\eta = W_B + 9/10 - \overline{d} \geq 0$ almost everywhere.

Proof. Write $k(s) = \rho(s) - 1/(2s)$ and $z = s/2$. Direct algebra gives $k(s) = [\operatorname{csch} z - 1/z + \operatorname{sech} z]/4$. For $0 < z \leq 3/4$, the power series imply

$$\frac{\sinh z}{z} \leq \frac{1}{1 - z^2/6}, \qquad \cosh z \geq 1 + z^2/2.$$

The first inequality follows from $(2j+1)! \geq 6^j$. For $b = z^2 \leq 9/16$,

$$(1 + b/2)(1 - b/6)^2 - 1 = \frac{b(12 - 10b + b^2)}{72} > 0.$$

Hence $z^2 \cosh z > \sinh^2 z$, so $\operatorname{csch} z - 1/z$ is decreasing; so is $\operatorname{sech} z$. Thus k decreases for $0 < s \leq 3/2$. Since $R' = -\rho$, for $0 < x < B$,

$$V'(x) = \rho(B - x) - \rho(B + x) \geq \frac{1}{2(B - x)} - \frac{1}{2(B + x)} = \frac{x}{B^2 - x^2} = W_B'(x).$$

Integration from zero and evenness prove the first assertion.

Only the large reserve rows require additional estimates. Since $t > 69/100$ and $u > 109/100$, (4.1) gives

$$W_B(2t - B) > \tfrac{1}{2}\log(625/184) > \frac{441}{809} + \frac{1}{3}\left(\frac{441}{809}\right)^3 > \frac{1063}{1800},$$
$$W_B(u - B) > \tfrac{1}{2}(34/75)^2 = \frac{578}{5625} > \frac{1}{10}.$$

The last finite comparisons are rational inequalities after clearing positive denominators. Seven t vertices have $|x| \geq 2t - B$, and seven v vertices and path interior vertices have $|x| \geq u - B$. The values of η are consequently

vertex type	η
seven center or w	W_B
seven t	$W_B - 1063/1800$
seven v	$W_B - 1/10$
path leaf	$W_B + 1/15$
path interior	$W_B - 1/20$
pair	$W_B + 2/5$.

Each is nonnegative on its physical band. □

Lemma 5.2 (Prime norm bound)**.** *The actual operator satisfies* $\|\mathcal{P}_B\| \leq 6/5$.

Proof. These weights are different from the reserve weights. On the seven chart choose $(5/6, 1, 2/3, 5/18)$ for the center, t, v, and w types. The coefficient bounds already proved give the following neighbor sums:

type	$\sum_{j\sim i} c_{ij} w_j$ upper bound	$(6/5)w_i$
center	1	1
t	6/5	6/5
v	71/90	4/5
w	1/3	1/3.

Apply $2c_{ij}|z_i z_j| \leq c_{ij}(w_j/w_i)|z_i|^2 + c_{ij}(w_i/w_j)|z_j|^2$ to each edge. For the path, equal weights give a maximal row below $23/20 < 6/5$; for the pair it is below $1/2$. The direct integral yields $|\langle f, \mathcal{P}_B f\rangle| \leq (6/5)\|f\|^2$ and hence the norm bound. □

Lemma 5.3 (A rational exterior surplus)**.** *The exact constants satisfy*

$$\delta_0 := V_0 - c_\Gamma + \frac{5}{2} > \frac{29}{14000}, \qquad j_7 := \frac{29}{5} - c_\Gamma > 0. \tag{5.1}$$

Proof. With $z = e^{-3/8}$, the two inverse trigonometric series combine to give $V_0 = 4\sum_{j\ge 0} z^{4j+1}/(4j+1)$. The alternating exponential polynomial through degree five gives

$$z > \sum_{j=0}^{5} \frac{(-3/8)^j}{j!} = \frac{900839}{1310720} > \frac{687}{1000} =: b.$$

The omitted terms in the combined series are positive, and the finite rational inequality

$$4\left(b + \frac{b^5}{5} + \frac{b^9}{9}\right) > \frac{5771}{2000}$$

gives $V_0 > 5771/2000$.

For the upper constant, the positive identity $22/7 - \pi = \int_0^1 x^4(1-x)^4/(1+x^2)\,\mathrm{d}x$ gives $\pi < 22/7$. Since $\sum_{j=0}^{5}(23/20)^j/j! > 22/7$, also $\log \pi < 23/20$. Equation (4.1) gives

$$\frac{842}{1215} < \log 2 < \frac{23581}{34020} < \frac{347}{500}.$$

Writing $H_n = \sum_{k=1}^{n} 1/k$, a telescoping integral yields

$$H_n - \log n - \gamma_E = \sum_{k=n}^{\infty} \int_0^1 \frac{1-y}{(k+y)(k+1)}\,\mathrm{d}y > \frac{1}{2}\sum_{k=n}^{\infty}(k+1)^{-2} > \frac{1}{2(n+1)}.$$

As $H_{16} = 2436559/720720$, it follows that

$$\gamma_E < \frac{2436559}{720720} - 4\frac{842}{1215} - \frac{1}{34} < \frac{29}{50}.$$

These finite rational inequalities are verified by clearing denominators; their formulas, rather than a prior computation's status, are the evidence. Consequently

$$c_\Gamma < \frac{23}{20} + \frac{29}{50} + \frac{11}{7} + 3\frac{347}{500} = \frac{9421}{1750},$$

and

$$\delta_0 > \frac{5771}{2000} - \frac{9421}{1750} + \frac{5}{2} = \frac{29}{14000}.$$

The same upper bound for c_Γ gives $j_7 > 0$. □

Proposition 5.4 (Lower bound for the same auxiliary operator)**.** *For the operator* (3.6)*,*

$$A_b \ge \lambda_b \,\mathrm{Id}, \qquad \lambda_b = a + \frac{29\omega}{14000} > 0. \tag{5.2}$$

The bound concerns this auxiliary finite-interval operator only.

Proof. Put $\Delta_\Gamma = V - V_0 - W_B \ge 0$ and $J_7[f] = (7 - c_\Gamma)\|f\|^2 - \langle f, \mathcal{P}_B f\rangle$. Lemmas 5.2 and 5.3 imply $J_7[f] \ge j_7\|f\|^2$. The closed form $\mathfrak{a}_b$ of (3.6) has the exact decomposition

$$\begin{aligned} \mathfrak{a}_b[f] &= a\|f\|^2 + \int_\Omega p_\eta |f|^2\,\mathrm{d}x + \omega S_{\mathrm{graph}}[f] \\ &\quad + \omega\int_\Omega (\Delta_\Gamma + \delta_0)|f|^2\,\mathrm{d}x + \theta J_7[f], \\ p_\eta &= \omega(W_B + 9/10 - \overline{d}). \end{aligned} \tag{5.3}$$

Indeed $\overline{d}$ and W_B cancel, and $\omega + \theta = 1$ leaves the arithmetic term $-\langle f, \mathcal{P}_B f\rangle$ exactly once. The remaining constant is $-c_\Gamma + (17/5)\omega + 7\theta = \kappa_g$. Each nonconstant term in (5.3) is nonnegative by the preceding lemmas. Thus

$$\mathfrak{a}_b[f] \ge (a + \omega\delta_0 + \theta j_7)\|f\|^2 \ge \lambda_b\|f\|^2.$$

The bound extends over the closed form domain and gives the associated operator inequality. The contribution of δ_0 occurs once. □

The residual reserve relative to the multiplier $q_0 = a + p_\eta + \theta j_7$ is at least $29\omega/14000$. This is not a computed smallest eigenvalue. No separate full-high diagonal reserve, internal-kernel remainder, or whole-pole term has been added to A_b to obtain (5.2). The complete pole in a different trial objective remains a separate object.

6 The projected functional and the missing numerical suppliers

For the canonical source define

$$g = aF - E\alpha, \quad c_b = A_b^{-1} g, \quad m = E^*(p_\eta F), \quad M_b = \langle p_\eta F, P c_b \rangle = \langle Em, A_b^{-1} g \rangle. \tag{6.1}$$

The function $p_\eta F$ belongs to L^2 by logarithmic endpoint growth. Thus Theorem 2.1 applies with $\lambda = \lambda_b$ and $G = E^* A_b E$. It also applies with the weaker valid lower bound $\lambda = a$. For a fixed x_0, define

$$C = m^T x_0, \quad D = m^T(-\alpha - G x_0), \quad N_m \geq \|m\|, \quad N_r \geq \|g - A_b E x_0\|.$$

Certified full-domain enclosures of these scalars would give

$$M_b \in C + \frac{D}{2\lambda} + \left[-\frac{N_m N_r}{2\lambda}, \frac{N_m N_r}{2\lambda} \right], \tag{6.2}$$

with outward treatment of every input error. For a single fixed x_0, one need not store the whole matrix G: a certified Gx_0 or the direct scalar D suffices, together with the other suppliers.

6.1 Quantities that cannot substitute for these inputs

In the audited stored files, the vector named `chi` has target $E^* A_b F$, not $m = E^*(p_\eta F)$. The scalar named `H` in the form packet has target $\mathfrak{a}_b[F] + 2|\ell|^2$ for a separate whole-pole moment ℓ; it is not the 16×16 matrix G. The stored `beta` vector is a response projection $E^* c_b$; it is neither m nor a prescribed certified x_0. Fibre response centers of dimensions $7, 4, 2$ are not one global low-dimensional trial Ex_0. Other residual or observation Grams have different operators and bases.

In particular

$$\|E^* r_0\| = \|-\alpha - G x_0\| \leq \|r_0\|$$

is a lower bound on the full residual norm, not a replacement upper bound. The omitted high component $(\mathrm{Id} - P) r_0$ matters. An energy bound $G \leq C_{\mathrm{low}} I$ also does not imply $\|A_b E x_0\| \leq C_{\mathrm{low}} \|x_0\|$.

6.2 Scope of the data conclusion

Within the files identified in the accompanying source manifest, no certified m, Gx_0, full residual norm, or matching scalar suppliers C, D, N_m, N_r were found. The conclusion is limited to that inspection scope, not all other workspaces or possible data holdings. The earlier full independent response verification was incomplete; its stored projection intervals are not used here as certified numerical suppliers. The numerical result affected by transcribed beta endpoints is also excluded. No new source response or Gram calculation was performed to fill the gap.

The exact polynomial and coefficient data define the target; their byte identity does not establish the truth of a numerical enclosure. The analytic lower bound in Proposition 5.4 is proved above from the exact operator definition. It does not turn an uncertified response output into a certificate. At present (6.2) is a specified certificate format, not an evaluated sign result.

7 Data availability

The source definitions, reproducibility scripts and audit corrections are publicly available at the immutable source/data revision

https://github.com/serabing-hash/theta-weighted-certificate/tree/ed9ce2c814e2ef64da4ea0337ada8257bb8b0c8f/research/2026-10-13-direct-m-residual-enclosure-paper/.

The source/data commit is `ed9ce2c814e2ef64da4ea0337ada8257bb8b0c8f`. Its manifest and files were read back through anonymous public GitHub URLs and matched the local bytes before the final manuscript was compiled. This fixed link identifies the source/data deposit; it does not need to identify the later commit containing this PDF itself.

The final standalone TeX, PDF and a ZIP containing exactly those two files are deposited separately in the same repository directory. The repository is `https://github.com/serabing-hash/theta-weighted-certificate`. The source/data revision contains `source_definitions.json`, the scientific input manifest, the reproduction README, audit corrections, and the document/static verification scripts. Internal filesystem paths, Library identifiers and private transfer URLs are not part of this deposit.

Exact b_m and α strings are copied mechanically from the audited inputs, with scientific source hashes retained. The fixed-polynomial and alpha input hashes have prefixes `0d702949feba1632` and `7fb2fede946e3899`; the input manifest contains their complete hashes. The recovered source-definition text has hash prefix `e31b29a48d2b4dcd`. Historical source archives are not redistributed in this directory. The required analytic definitions and proofs are written in this manuscript, while the exact fixed coefficient strings are supplied with the public definitions. Hashes establish data identity, not proof by an earlier audit's PASS status.

The TeX file includes its bibliography internally and requires no external figure, bibliography, or coefficient input to compile. No earlier paper by the author is cited. The proof scope is the abstract residual estimate and the explicitly defined finite-interval auxiliary operator. A relationship to an original whole-line realization, a numerical sign for M_b, and any Riemann-hypothesis conclusion remain outside the established results.

References

[1] E. B. Davies, *Spectral Theory and Differential Operators*, Cambridge Studies in Advanced Mathematics 42, Cambridge University Press, 1995. doi:10.1017/CBO9780511623721. The spectral theorem is developed in Chapter 2, pp. 22–44.

[2] National Institute of Standards and Technology, *Digital Library of Mathematical Functions*, Section 18.3, "Definitions" (classical orthogonal polynomials), https://dlmf.nist.gov/18.3.